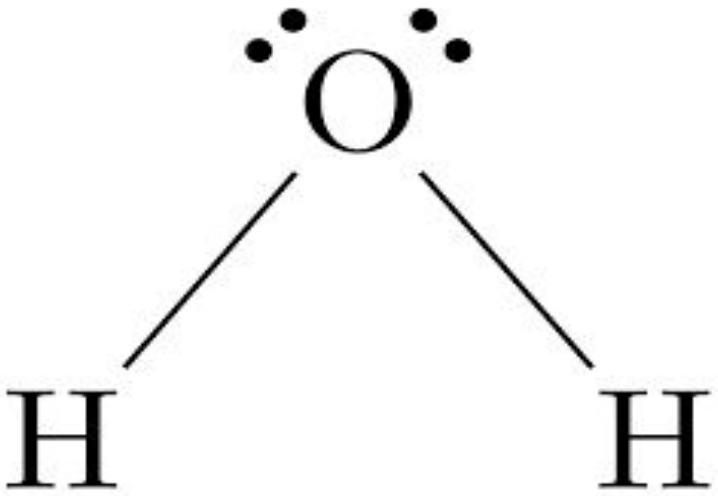
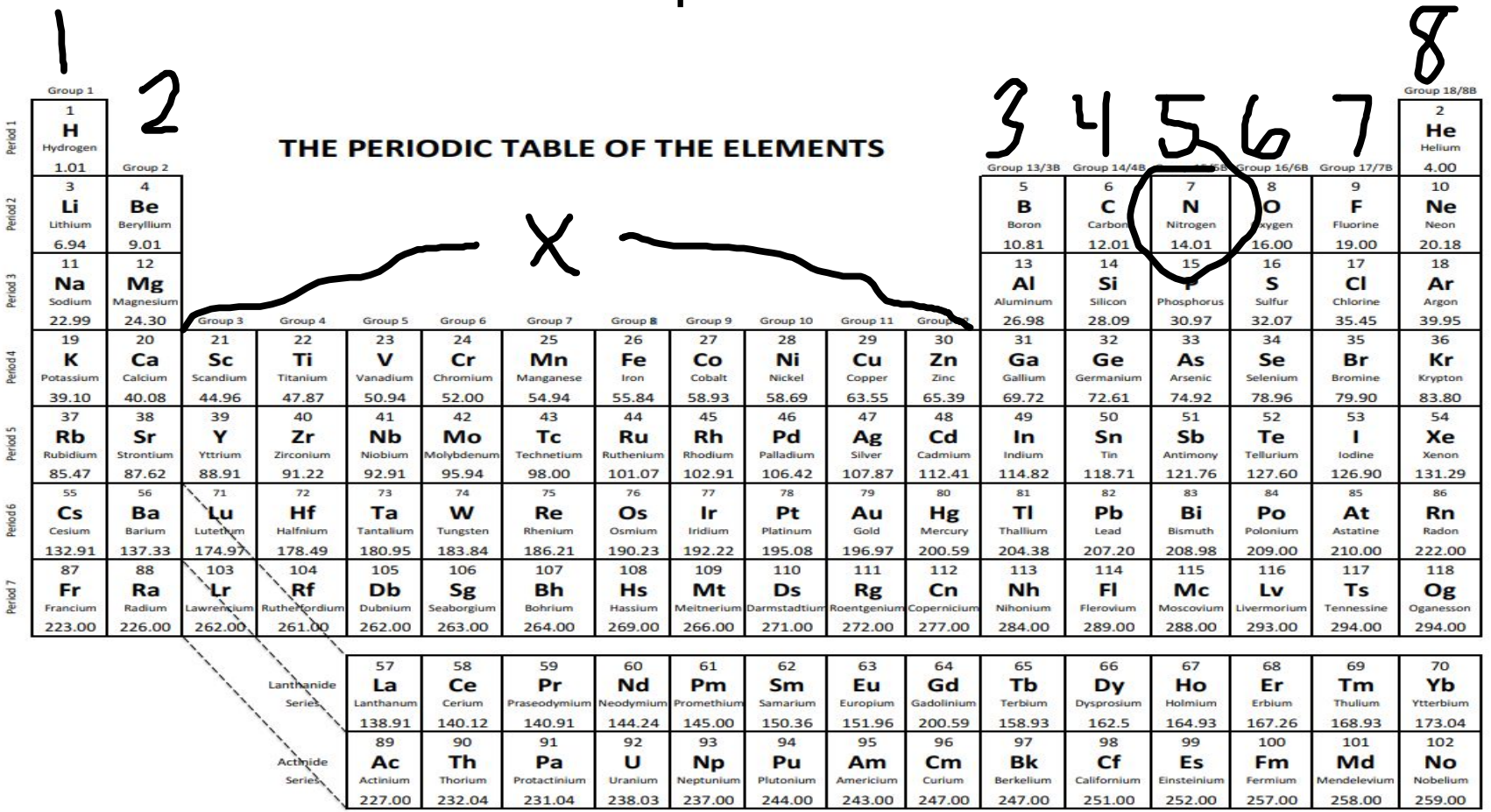
Lewis Structures



Valence electrons - electrons in the outermost energy level

Vertical Columns = Groups = # of Valence Electrons



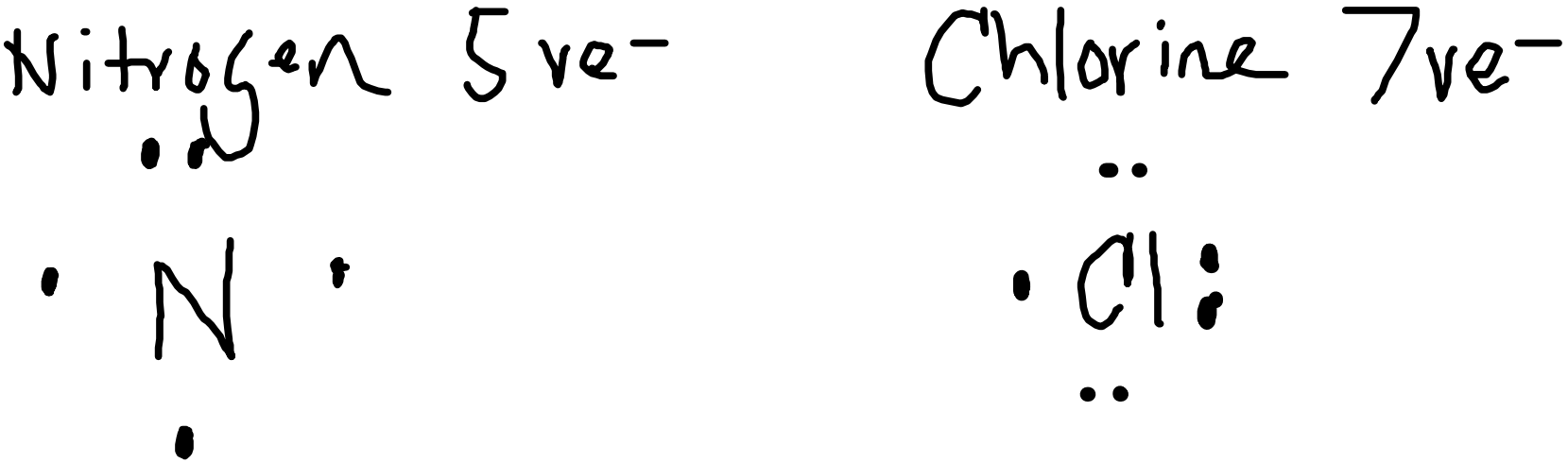
THE PERIODIC TABLE OF THE ELEMENTS

	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8	Group 9	Group 10	Group 11	Group 12	Group 13/3B	Group 14/4B	Group 15/5B	Group 16/6B	Group 17/7B	Group 18/8B
Period 1	1 H Hydrogen 1.01																	2 He Helium 4.00
Period 2	3 Li Lithium 6.94	4 Be Beryllium 9.01											5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18
Period 3	11 Na Sodium 22.99	12 Mg Magnesium 24.30											13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95
Period 4	19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.87	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.84	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.39	31 Ga Gallium 69.72	32 Ge Germanium 72.61	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80
Period 5	37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium 98.00	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
Period 6	55 Cs Cesium 132.91	56 Ba Barium 137.33	71 Lu Lutetium 174.97	72 Hf Halfnium 178.49	73 Ta Tantalium 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.20	83 Bi Bismuth 208.98	84 Po Polonium 209.00	85 At Astatine 210.00	86 Rn Radon 222.00
Period 7	87 Fr Francium 223.00	88 Ra Radium 226.00	103 Lr Lawrencium 262.00	104 Rf Rutherfordium 261.00	105 Db Dubnium 262.00	106 Sg Seaborgium 263.00	107 Bh Bohrium 264.00	108 Hs Hassium 269.00	109 Mt Meitnerium 266.00	110 Ds Darmstadtium 271.00	111 Rg Roentgenium 272.00	112 Cn Copernicium 277.00	113 Nh Nihonium 284.00	114 Fl Flerovium 289.00	115 Mc Moscovium 288.00	116 Lv Livermorium 293.00	117 Ts Tennessine 294.00	118 Og Oganesson 294.00
Lanthanide Series			57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium 145.00	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 200.59	65 Tb Terbium 158.93	66 Dy Dysprosium 162.5	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.04
Actinide Series			89 Ac Actinium 227.00	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium 237.00	94 Pu Plutonium 244.00	95 Am Americium 243.00	96 Cm Curium 247.00	97 Bk Berkelium 247.00	98 Cf Californium 251.00	99 Es Einsteinium 252.00	100 Fm Fermium 257.00	101 Md Mendelevium 258.00	102 No Nobelium 259.00

Lewis Dot Structures

Steps…

1. Determine how many valence electrons the atom has
2. Write the symbol of the atom
3. Place an electron on each “side” of the atom until all valence electrons are distributed. (no more than two per “side”)



Lewis Structures

- Visual way of displaying the electron arrangement throughout the structure of the molecule
- Follows the octet rule
  - Octet rule - the tendency of atoms to have 8 valence electrons
  - If atoms do not have 8 valence electrons, they form bonds with other atoms to meet the requirement.
  - Atoms needs 8 valence electrons (exception is Hydrogen which needs 2)

Lewis Structure

- Steps…
  - Find the total # of valence electrons (adjust for ions)
  - Arrange atoms around the central atom
  - Form bonds (line = 2 electrons) between the central atom and outside atom(s)
  - Distribute remaining atoms (outside atoms first) Remember the octet rule
  - If there aren't enough atoms to go around, form double or triple bonds

Lewis Structure - CF_4

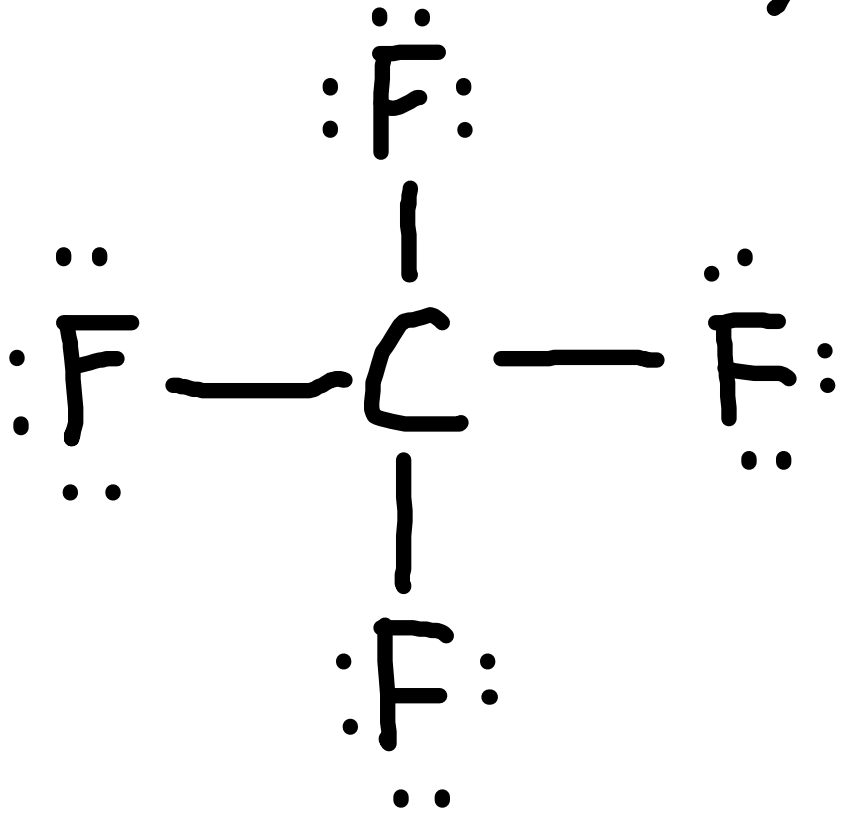
$4 + 4(7) = 32$ ve⁻

-8

24

-24

0



Lewis Structure - CO_2



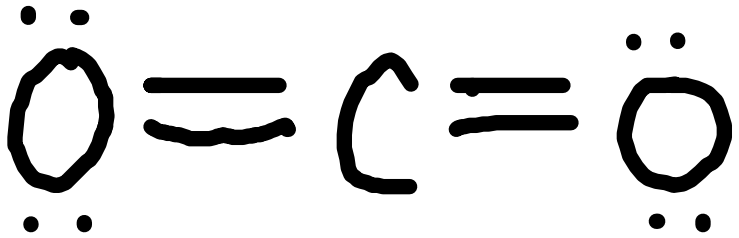
$4 + 2(6) = 16 \text{ ve}^-$

-4

12

-12

$\emptyset$



Lewis Structure - HCN

$1 + 4 + 5 = 10$ vc$^-$

-4

6



-6

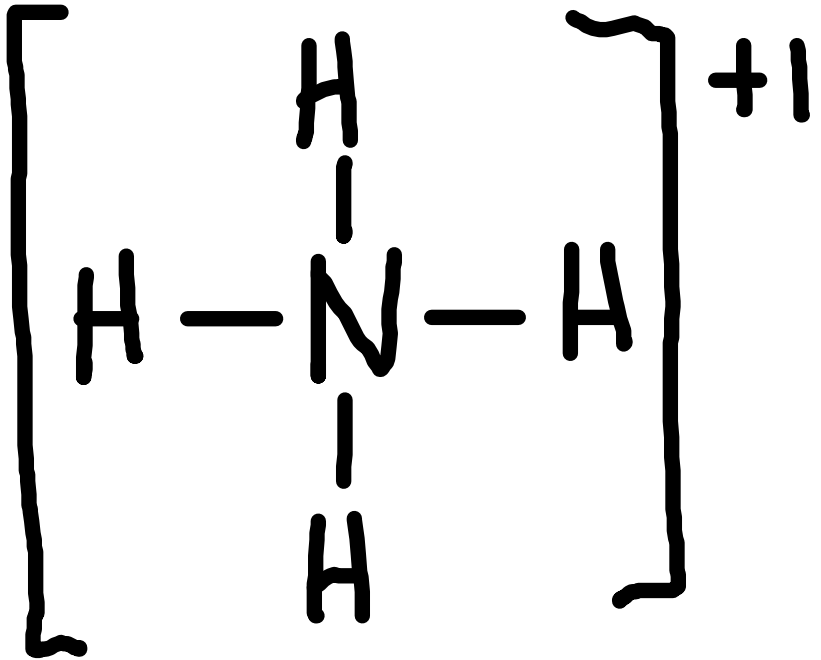
0

Lewis Structure - $[NH_4]^{+1}$

$5+4(1) = 9-1 = 8\,v.e^-$

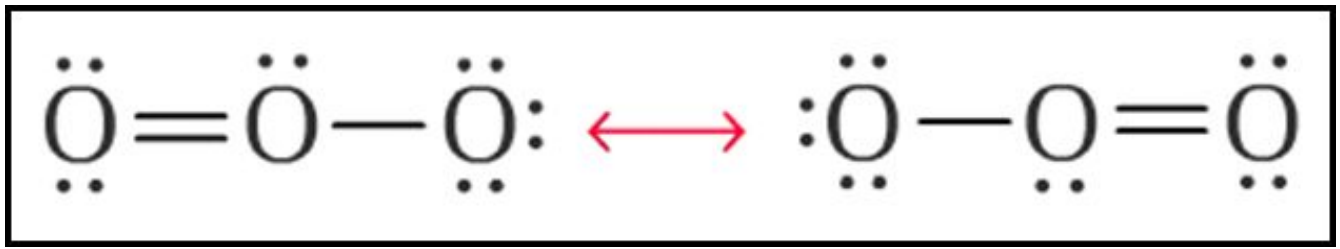
-8

$\emptyset$



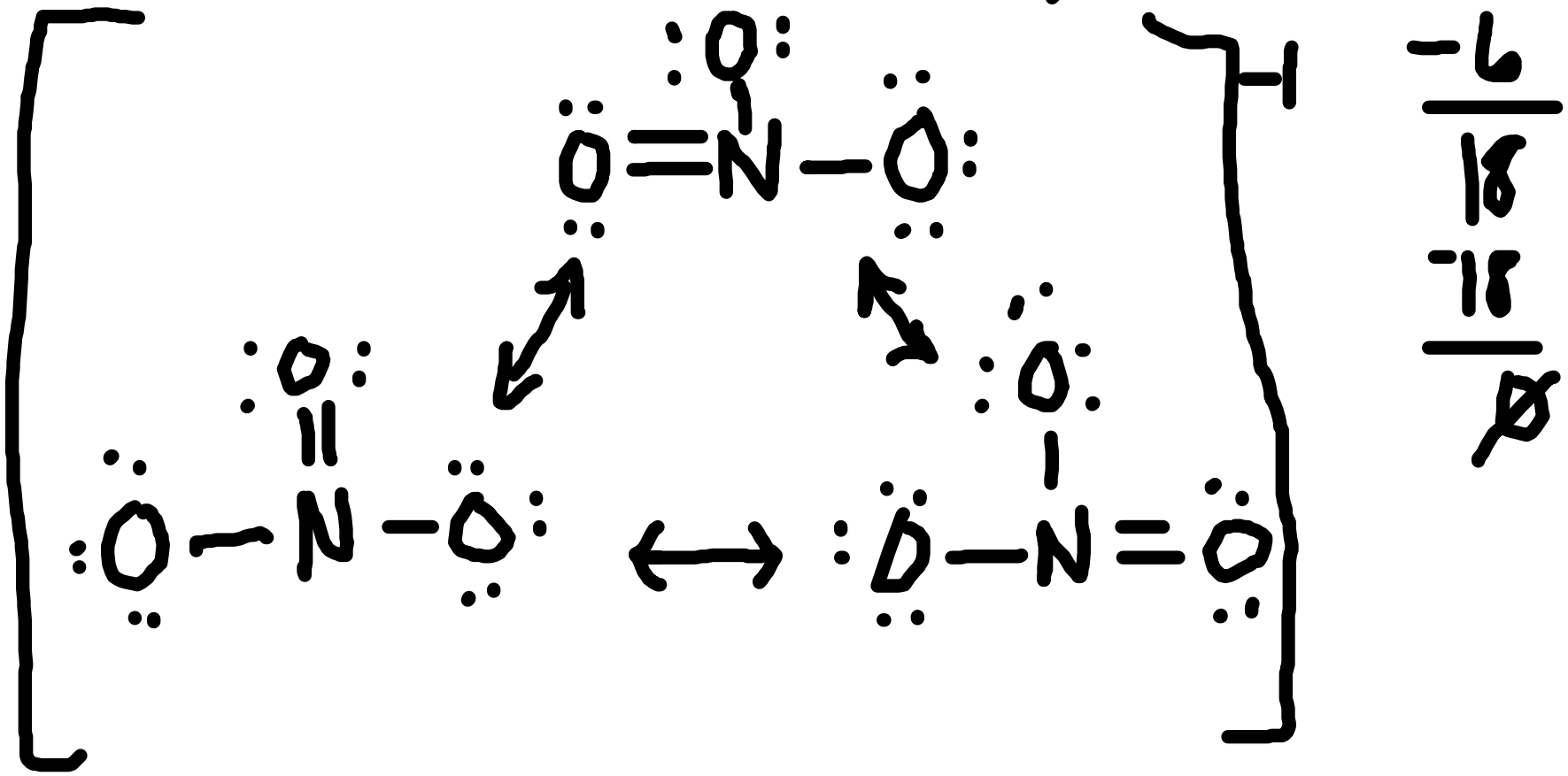
Resonance Structures

- Molecules that cannot be correctly represented by a single Lewis Structure
- Actual structure is an average of all the possibilities
- Show all possible structures separated with a double headed arrow



Lewis Structure - $[NO_3]^{-1}$

$5+3(6)+1=24$ ve⁻



-6

18

-18

0